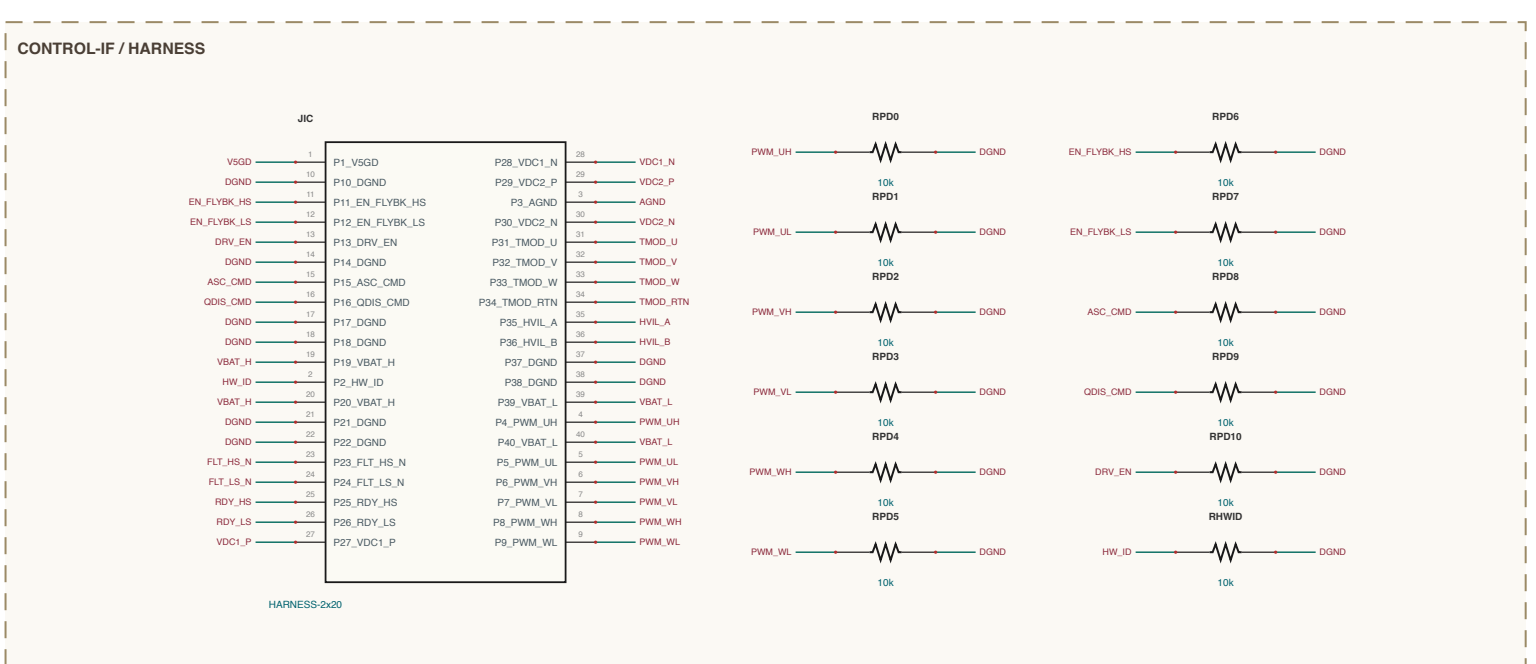
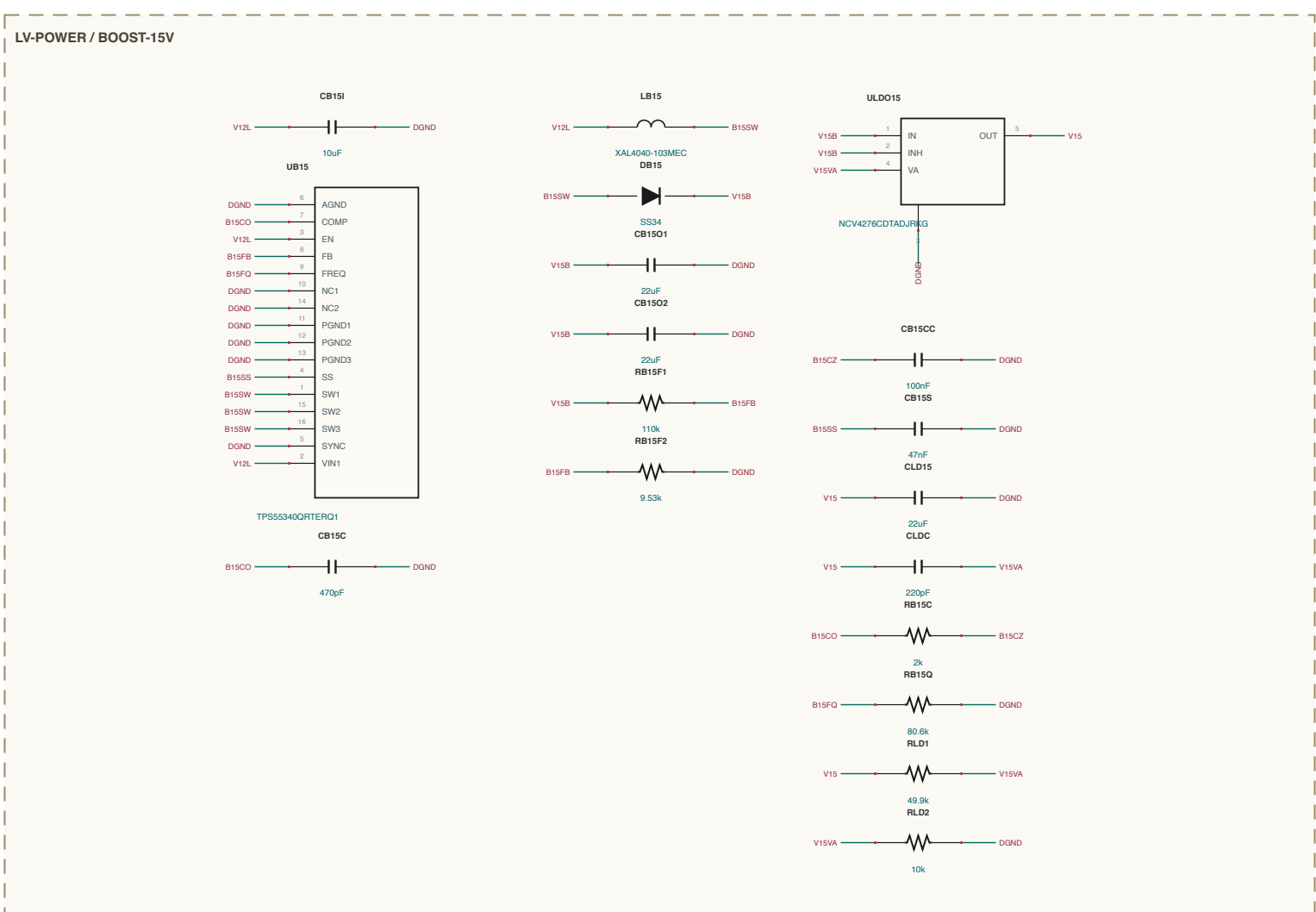
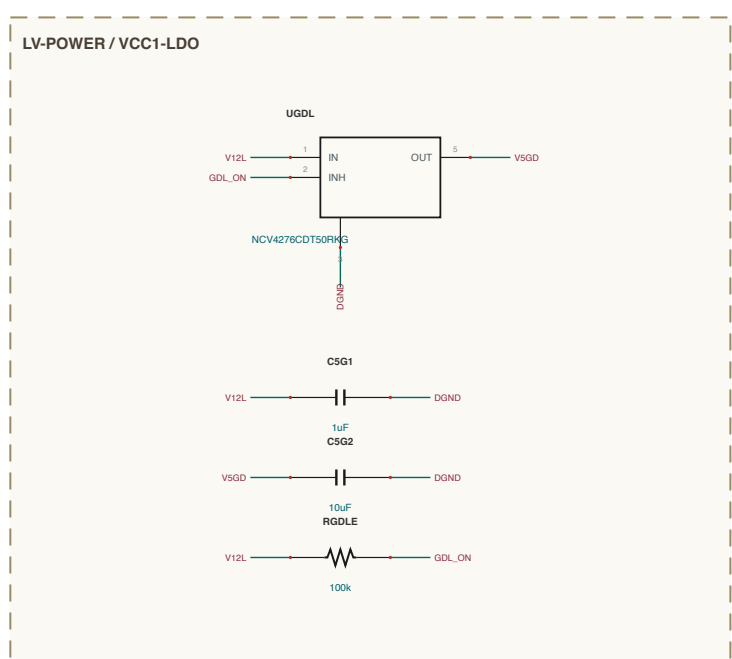
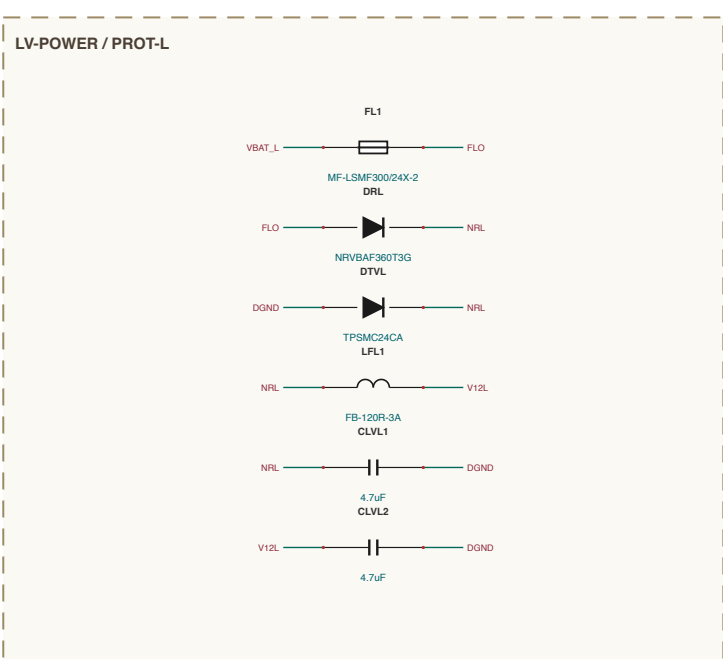
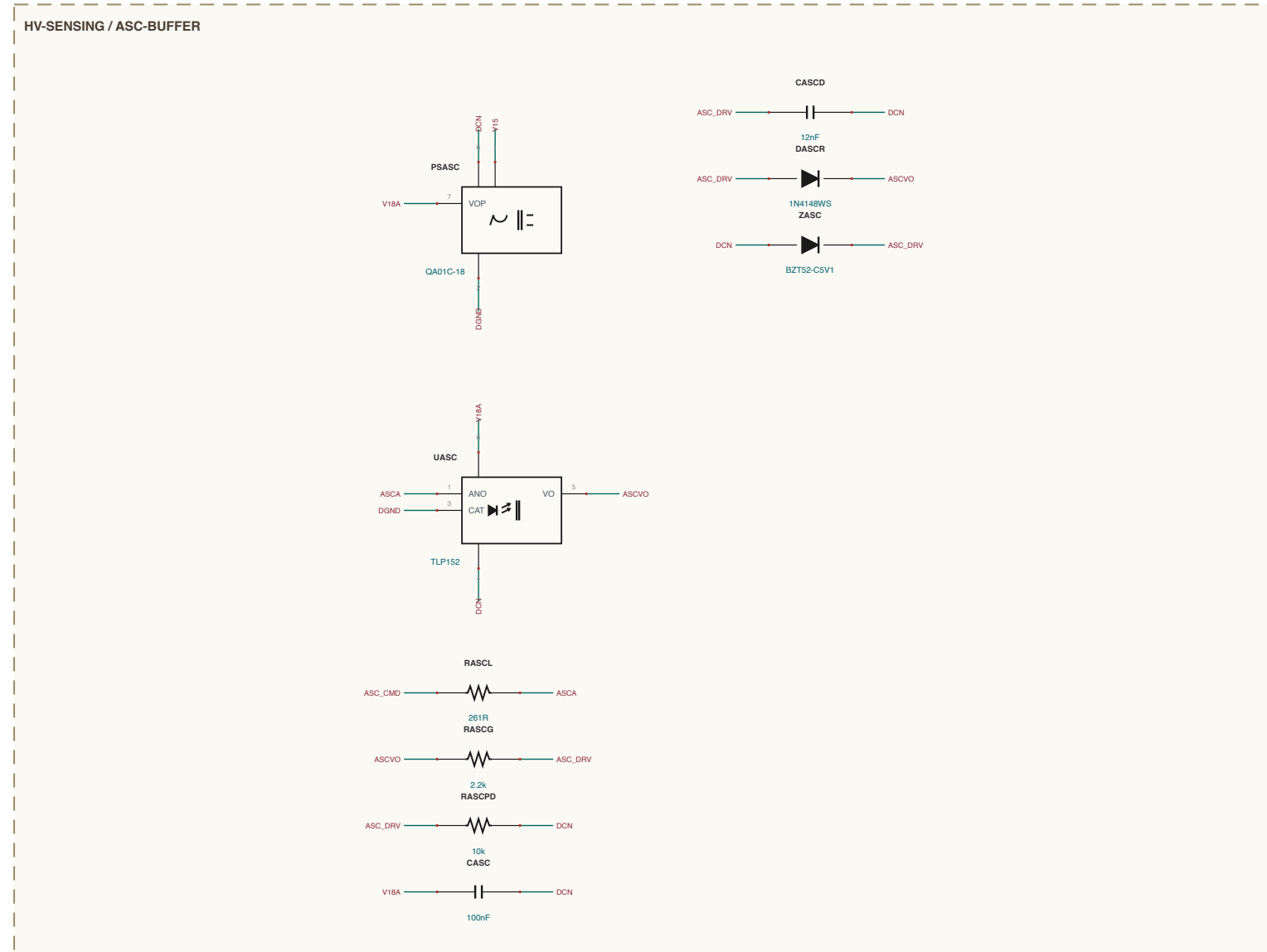
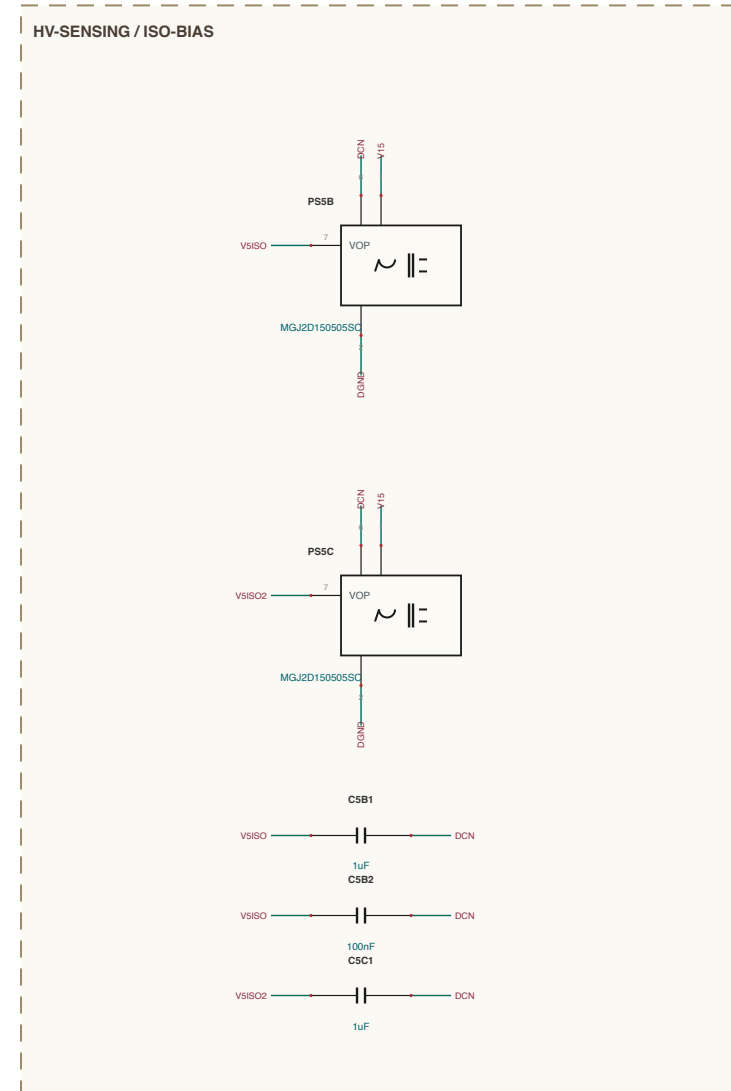
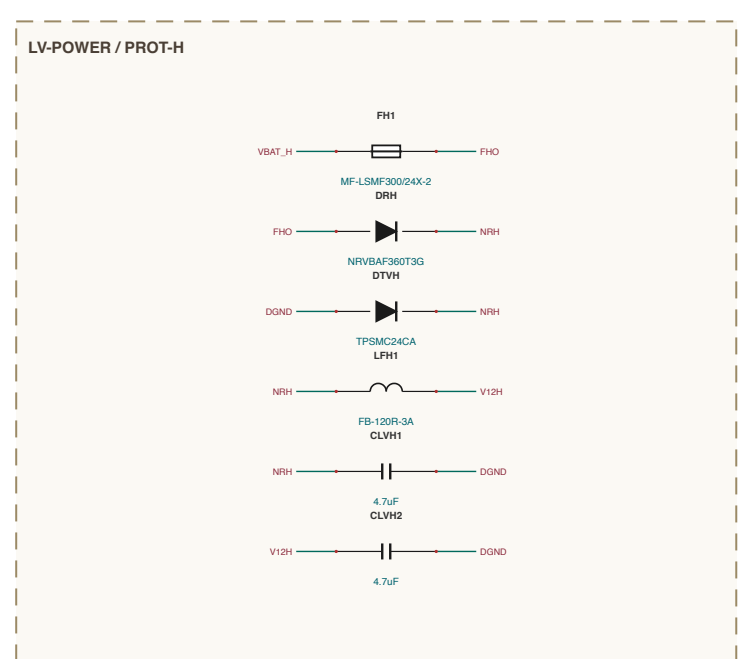
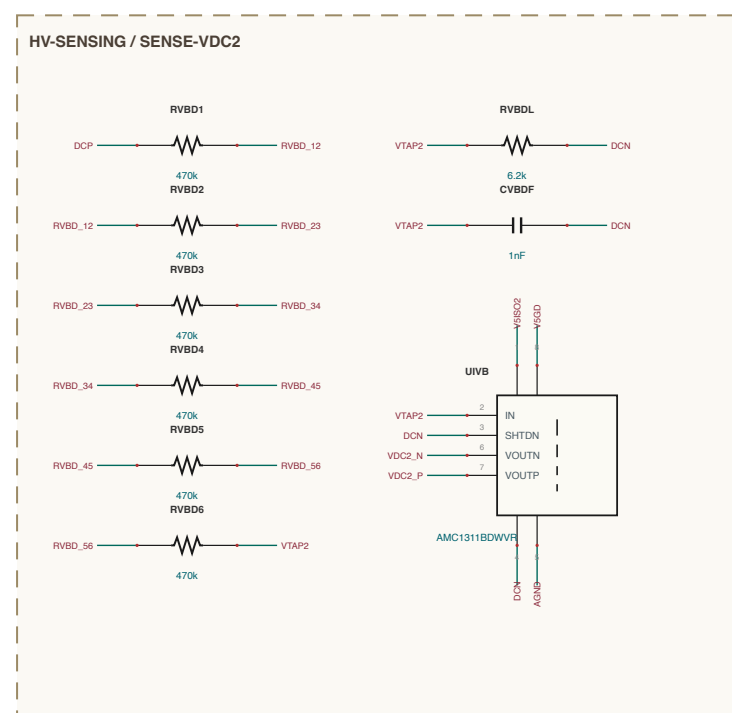
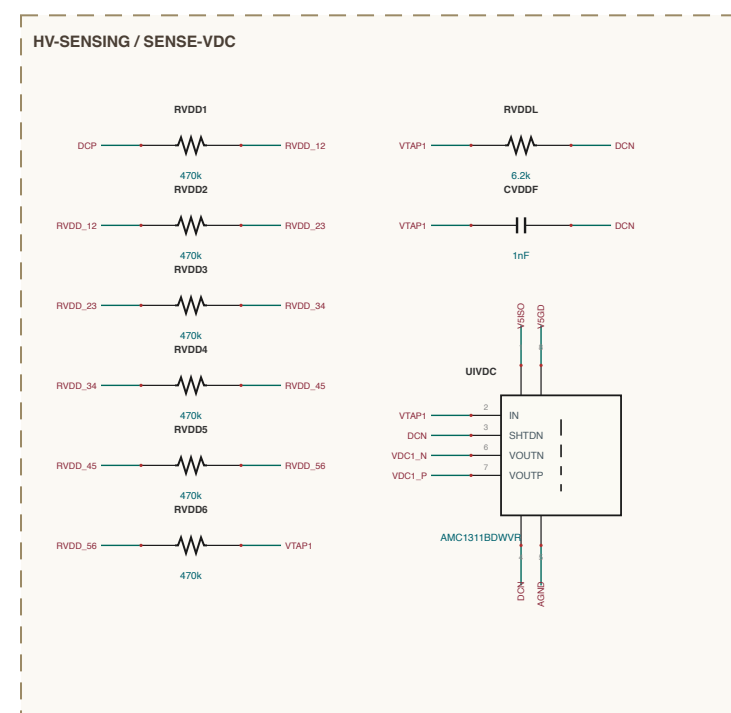
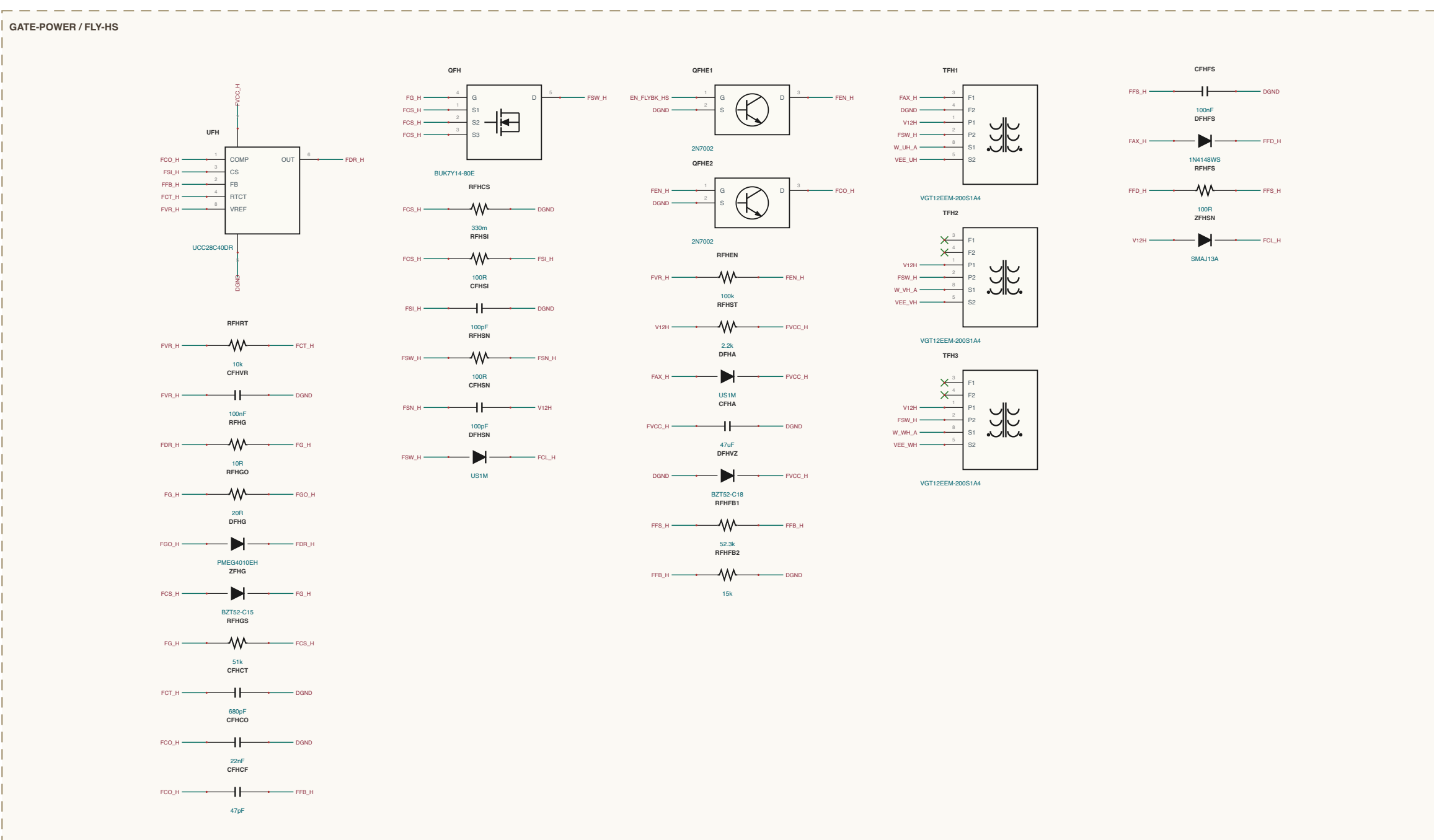
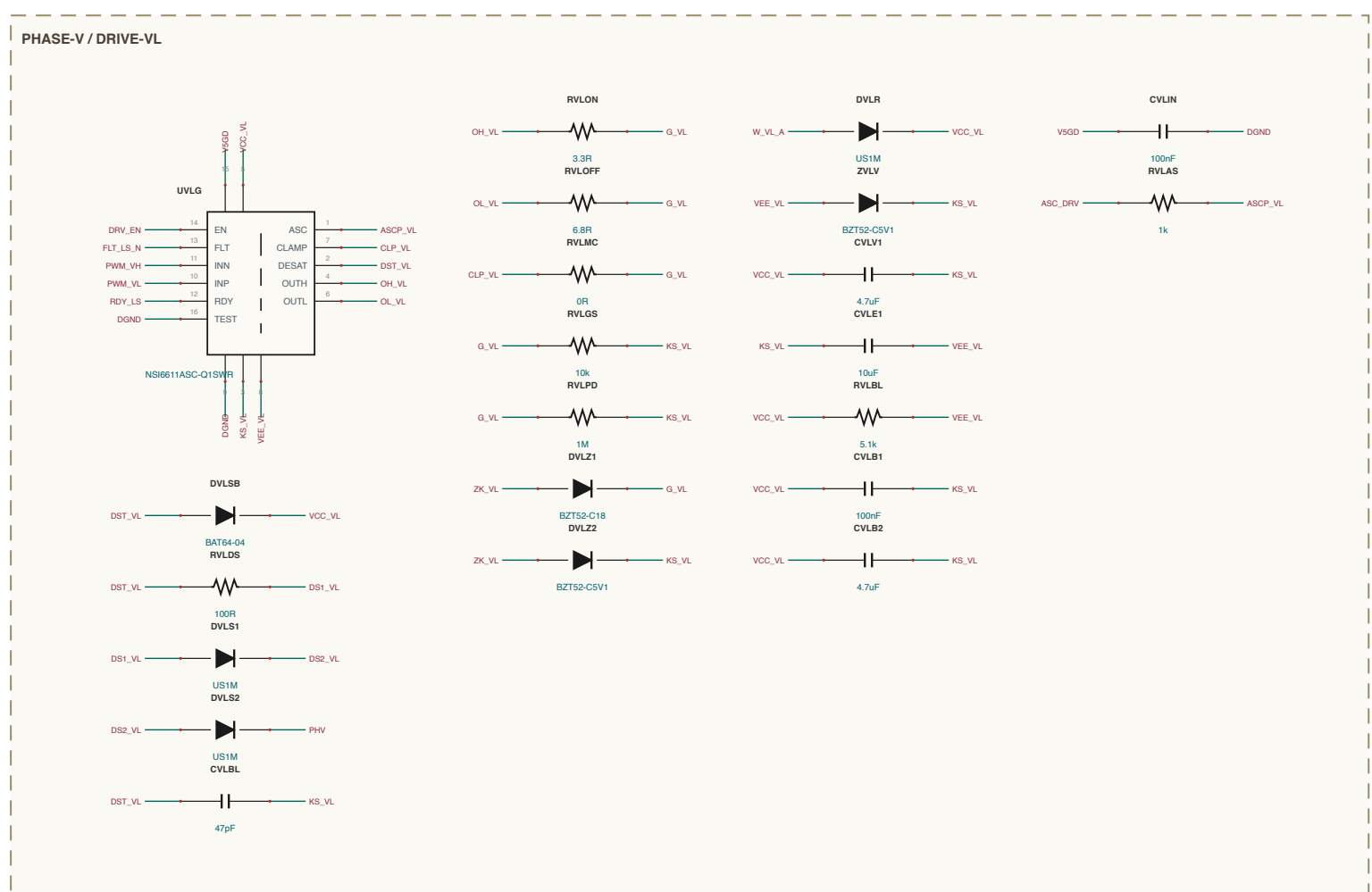
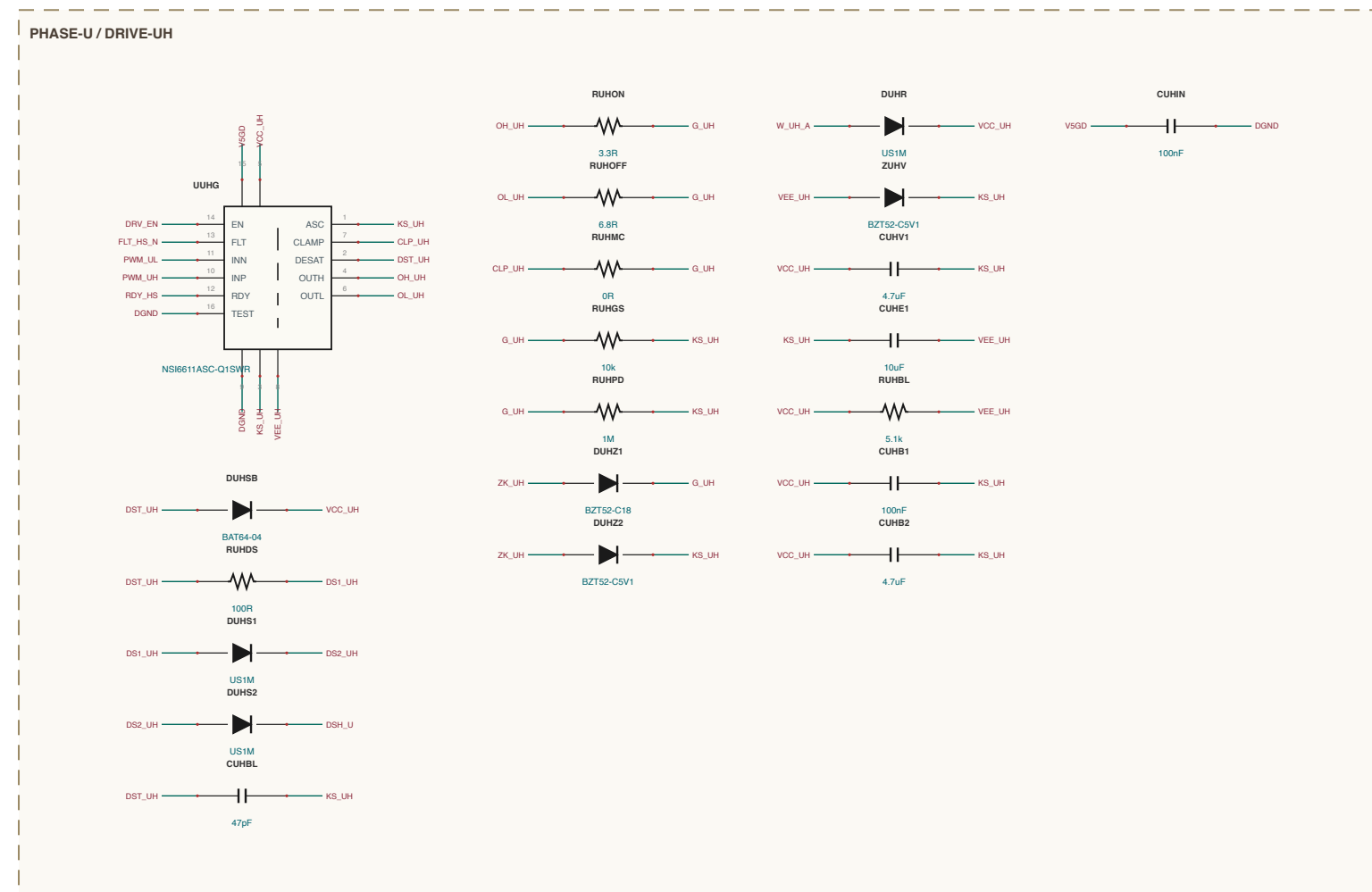
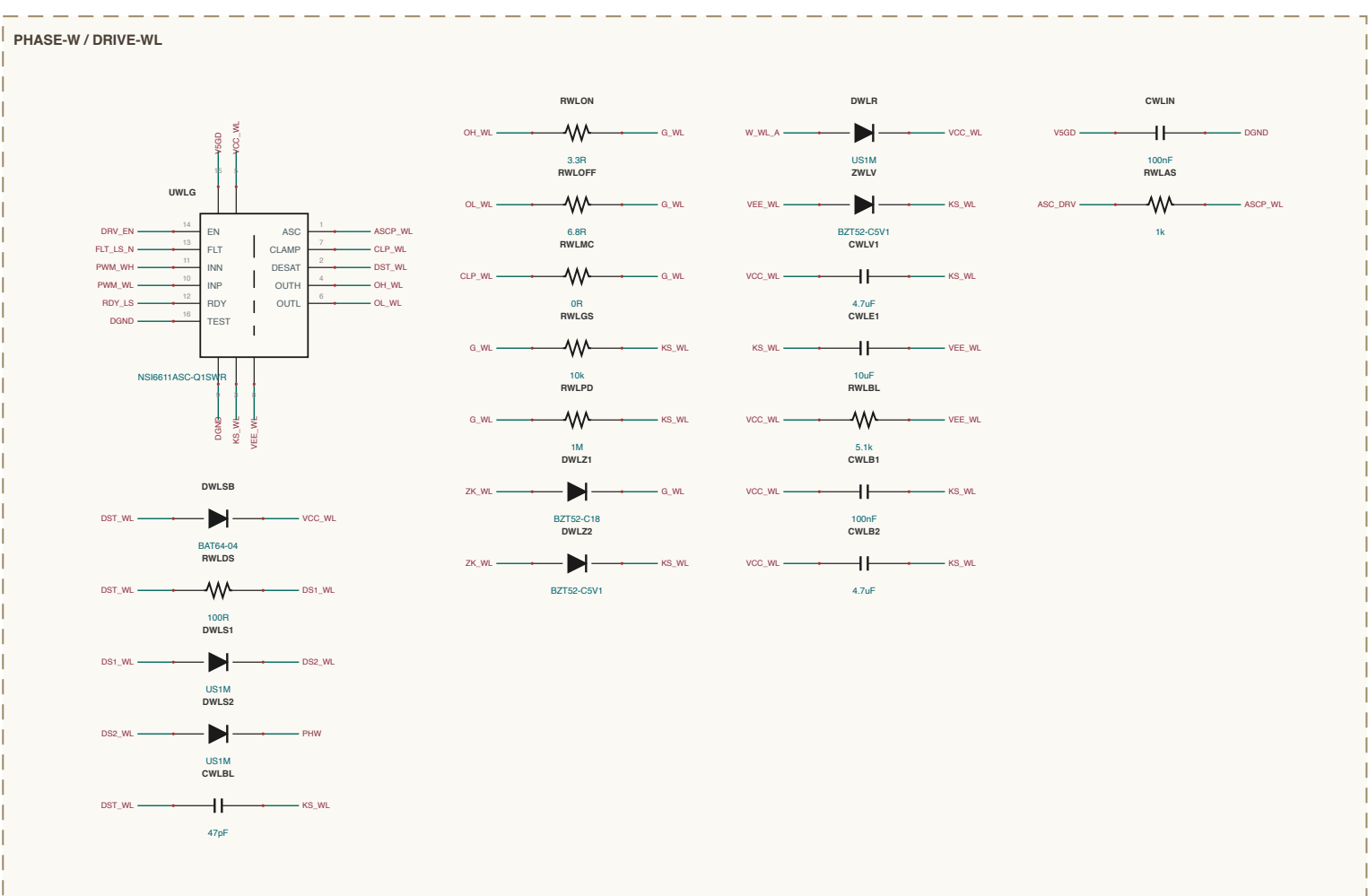
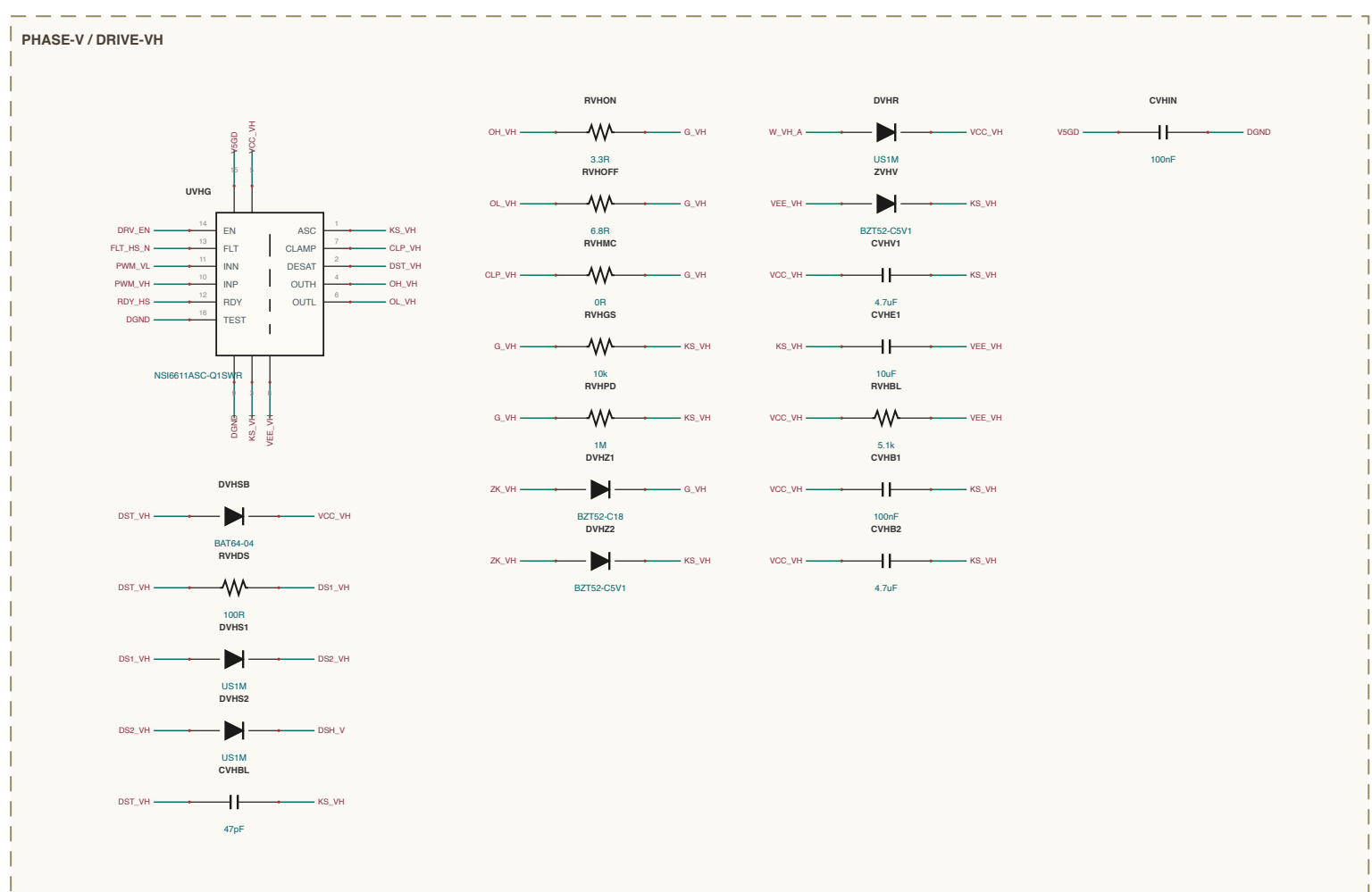
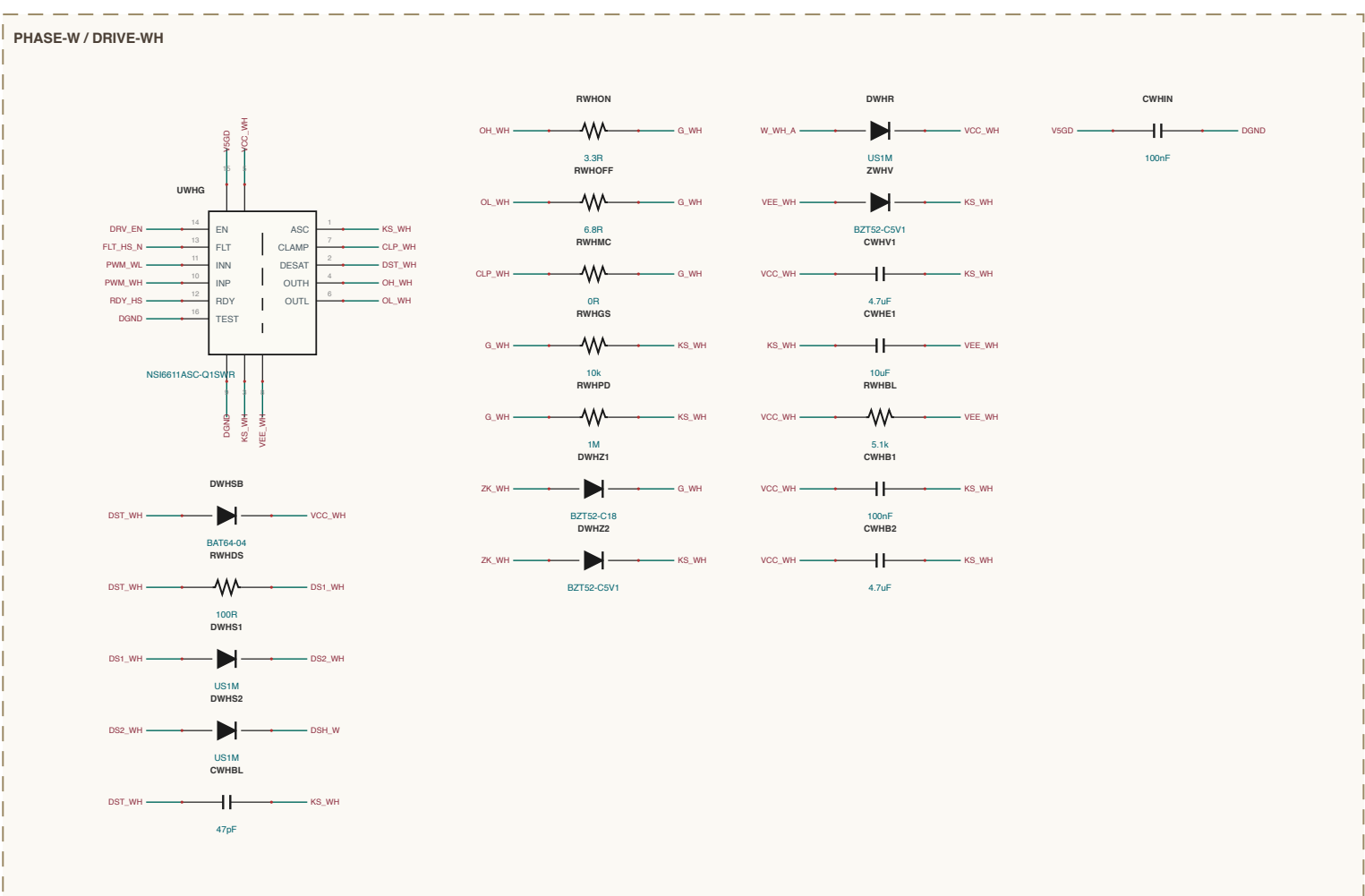
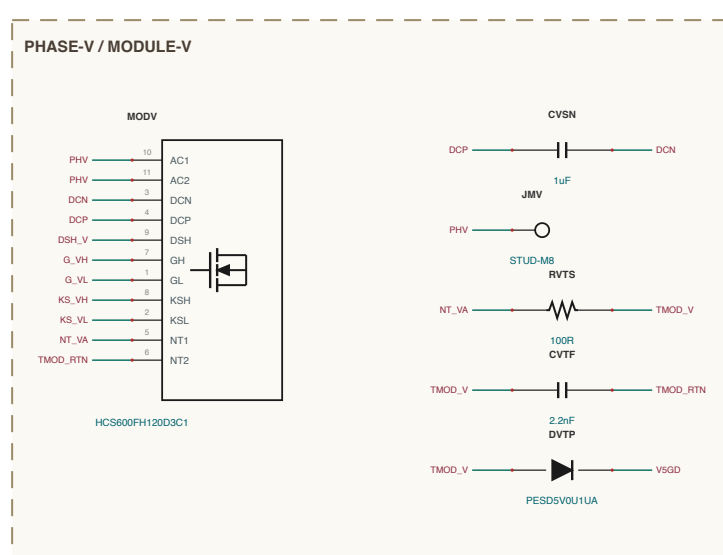
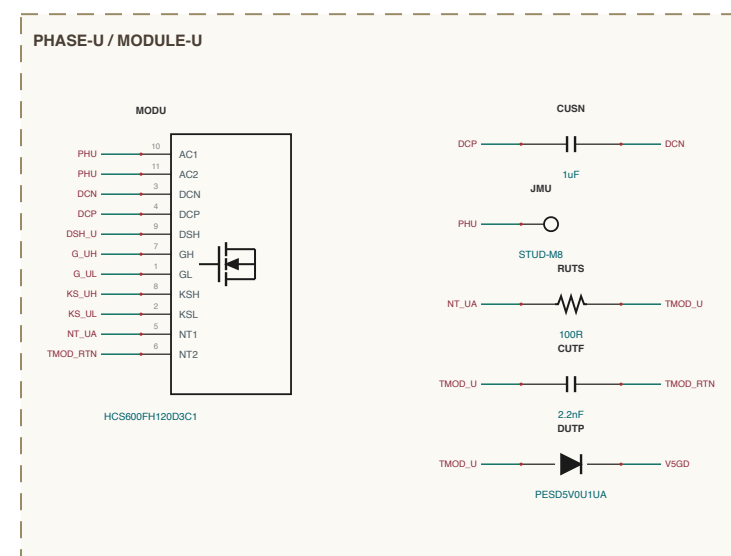
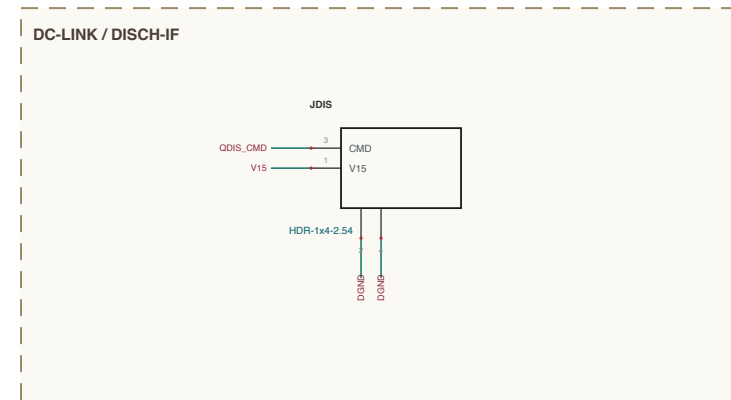
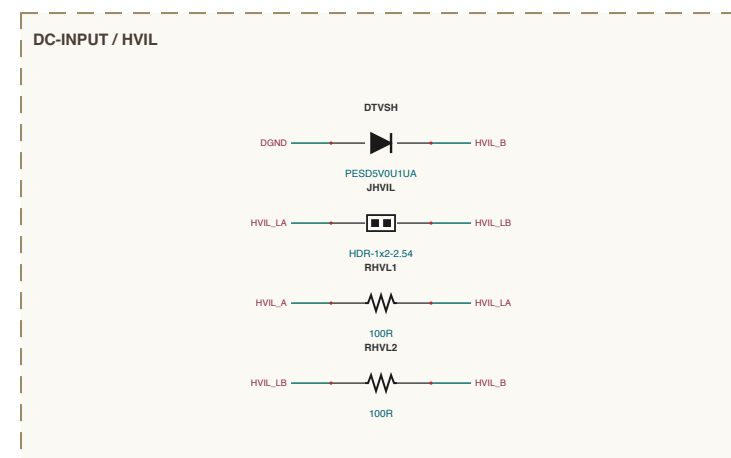
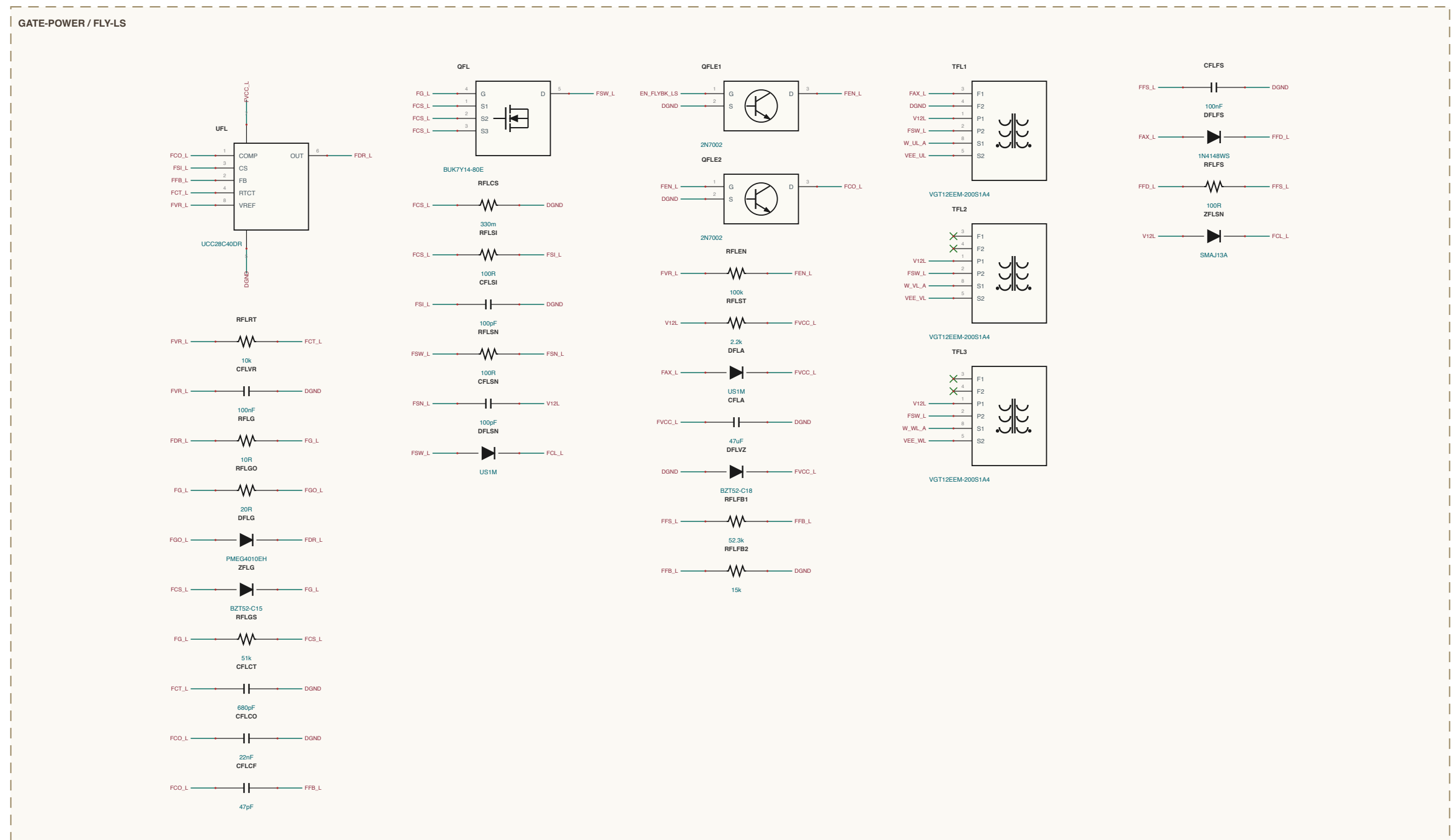
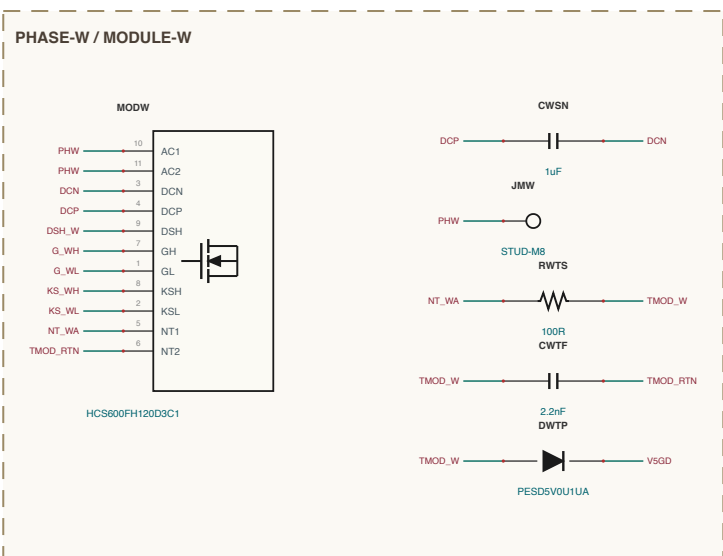
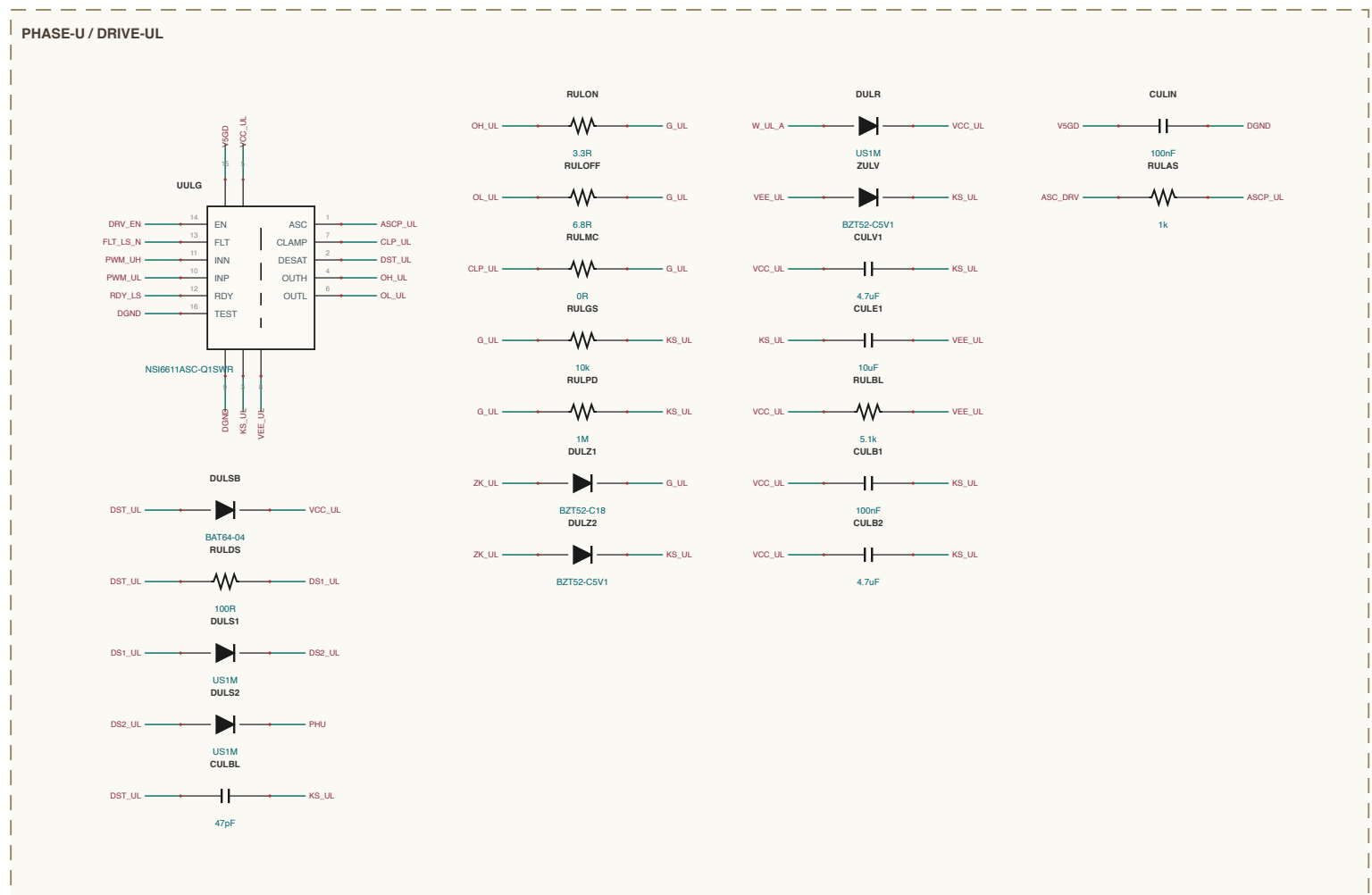
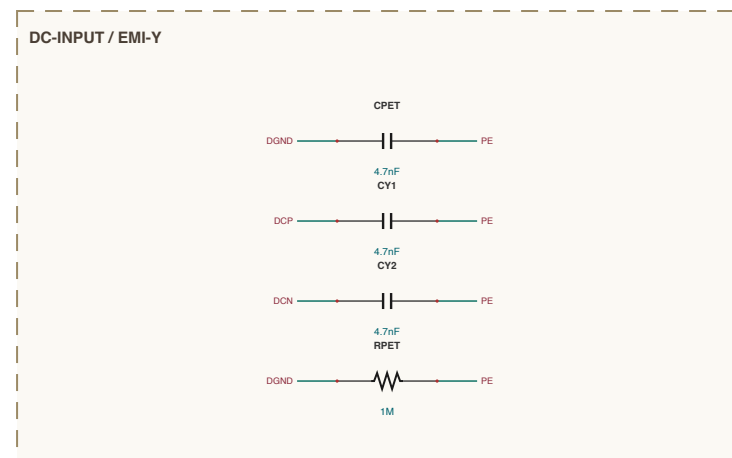
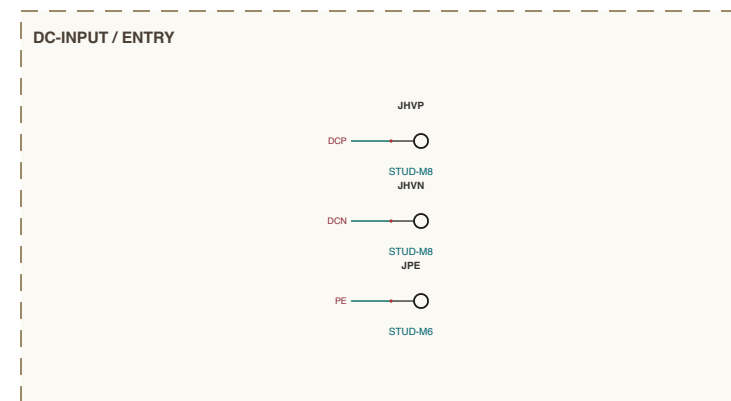




SHEET INDEX

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HYBRIDING - 4 sections

1-2 POWER - 4 sections

DC-BUSBT - 3 sections

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PHASE-W - 3 sections

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GATE-POWER - 2 sections

CONTROL_UF - 1 section

DC-LINK - 1 section

see A.3 - 16 sections - 398 components

NET NAMING

Internal functions are named for what they *do*

Unit- or File- mode at that IC pin e.g. `GPIOC_WRIP`

Pin- or signal- id of a socket pin e.g. `RELA_A`

Pin- or -io- id between `Proteus` and e.g. `W5502_12`

All others are English design terms

Name same as another chip + same ref. joined only at the end

Joined `connectors` to `UnitId` (see `INTERCONNECT` `UNITs`, design `units`)

ASL-0 SAFETY CONCEPT + DC-LINK SAFETY VERIFICATION

Safe state = 3 phase open (ASL does overcurrent), discharge via ECR 802/ 805 kA/s

1 Locking S20206 (ASL-0 con) turn torque path

2 ECR ASL switching – monitors → F058 (1s) F06

3 F058BMCU_UNMSTRM15.7 fault → 20V, SW = high open

4 F058 stop → LB ASL (with F058 stopped OVF)

5 Rydam SW + BMCU DR F058-GPH1 (ASL gate power)

6 1s to fault + 3s(2.0s) off-availability (torque path)

7 2s to low IOC sensor v_{DC} = fault, F058 monitored

8 DEAR? per switch (2s + 1s on delay), latching

9 SW open → VDDSDI0 degradation → torque off (F06)

— DISCHARGE —

PASSIVE 1s (2s DISCHG 10s + 2s + 80s) 5s + 10 s to 80 V

ACTIVE 0.05s + 1s DISCHG 10s + 1.0s + 1.6 s to 80 V

E = 105 ± 105A = 32.2 J/m.s worst: 43k, 2000V/s

VBAT? IBCU feeds GND, switches BOTH IOC channels

NOTE: never merge with the DISCHARGE BOARD Break